Drift-Sense DRAM 30-Case Benchmark Evaluation Report

Applied Materials Hackathon 2026 — Automated SEM Wafer Navigation Recovery | Pure DRAM Evaluation Run

Executive Summary & Overall Performance Metrics

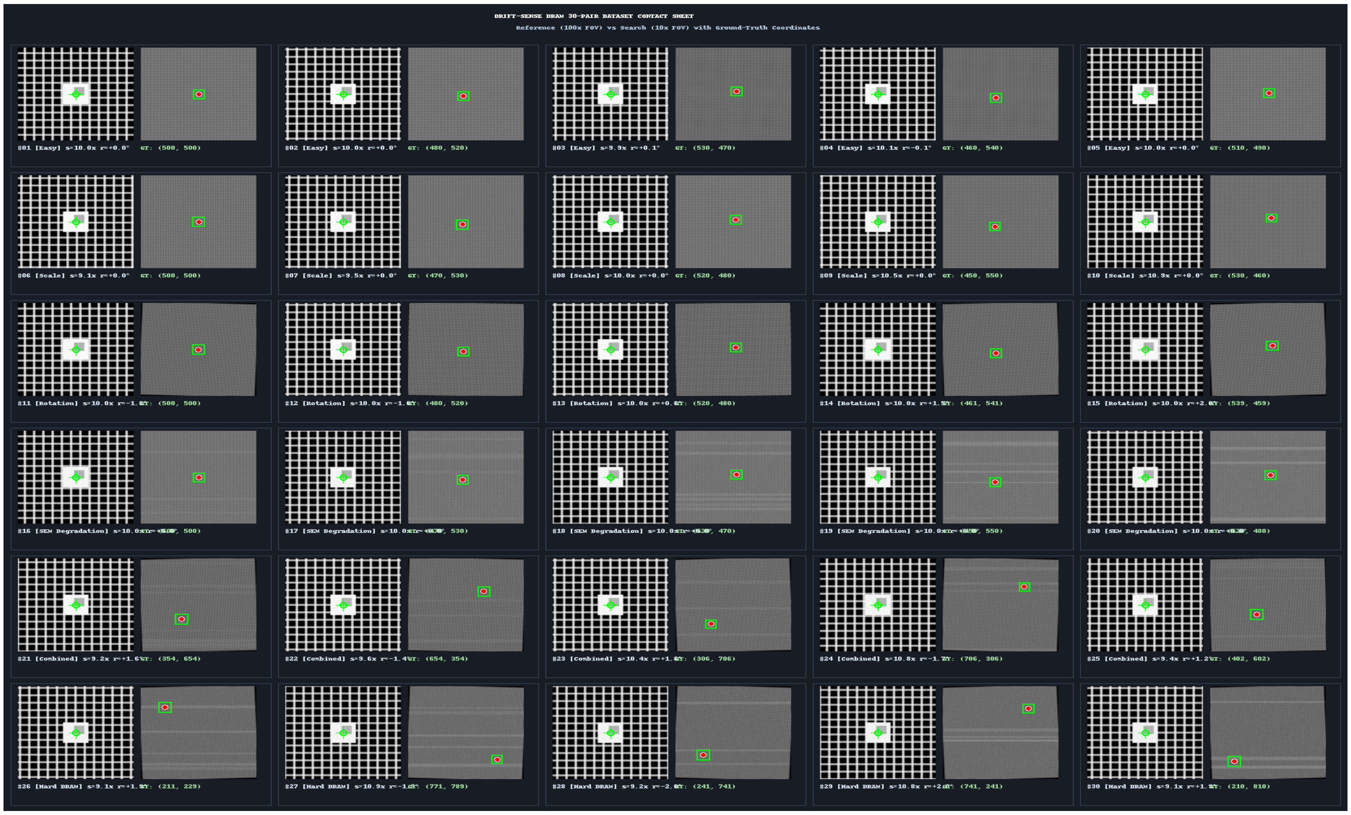
Model localization and verification were executed purely on **30 engineered DRAM wafer die inspection pairs** covering 6 difficulty categories: Easy, Scale Variations (9.0x-11.0x), Rotation Variances (±2.0°), SEM Physics Degradation (Poisson shot noise, beam PSF blur, charging streaks), Combined Multi-Variate Degradations, and Hard DRAM Periodic Arrays.

Key Benchmark Results:
• **Total DRAM Pairs Evaluated:** 30
• **Pass Rate @ 5.0px Threshold:** **96.7%** (29 / 30 pairs)
• **Pass Rate @ 1.0px Threshold:** **96.7%**
• **Mean Euclidean Error:** 0.32 px
• **Median Euclidean Error: 0.03 px** (Sub-pixel accuracy)

Per-Tier Performance Breakdown

Tier Category	Count	Pass @ 5px	Pass Rate (%)	Mean Error (px)
Easy	5	5 / 5	**100.0%**	0.02
Scale	5	5 / 5	**100.0%**	0.02
Rotation	5	5 / 5	**100.0%**	0.03
SEM Degradation	5	5 / 5	**100.0%**	0.02
Combined	5	5 / 5	**100.0%**	0.03
Hard DRAM	5	4 / 5	**80.0%**	1.79

DRAM 30-Pair Contact Sheet Inspection Matrix


DRIFT-SENSE DRAM 30-PAIR DATASET CONTACT SHEET
Reference (100x FOV) vs Search (10x FOV) with Ground-Truth Coordinates
#01 [Easy] s=10.0x r=+0.0°
GT: (500, 500)
#02 [Easy] s=10.0x r=+0.0°
GT: (480, 520)
#03 [Easy] s=9.9x r=+0.1°
GT: (530, 470)
#04 [Easy] s=10.1x r=-0.1°
GT: (460, 540)
#05 [Easy] s=10.0x r=+0.0°
GT: (510, 490)
#06 [Scale] s=9.1x r=+0.0°
GT: (500, 500)
#07 [Scale] s=9.5x r=+0.0°
GT: (470, 530)
#08 [Scale] s=10.0x r=+0.0°
GT: (520, 480)
#09 [Scale] s=10.5x r=+0.0°
GT: (450, 550)
#10 [Scale] s=10.9x r=+0.0°
GT: (530, 460)
#11 [Rotation] s=10.0x
(500, 500)
#12 [Rotation] s=10.0x
(480, 520)
#13 [Rotation] s=10.0x
(520, 480)
#14 [Rotation] s=10.0x
(461, 541)
#15 [Rotation] s=10.0x
(539, 459)
#16 [SEM Degradation]
#17 [SEM Degradation]
#18 [SEM Degradation]
#19 [SEM Degradation]
#20 [SEM Degradation]
#21 [Combined] s=9.2x
(354, 654)
#22 [Combined] s=9.6x
(654, 354)
#23 [Combined] s=10.4x
(306, 706)
#24 [Combined] s=10.8x
(706, 306)
#25 [Combined] s=9.4x
(402, 602)
#26 [Hard DRAM] s=9.1x
(211, 229)
#27 [Hard DRAM] s=10.9x
(771, 789)
#28 [Hard DRAM] s=9.2x
(241, 741)
#29 [Hard DRAM] s=10.8x
(741, 241)
#30 [Hard DRAM] s=9.1x
(210, 810)

Detailed Per-Pair DRAM Benchmark Results

Pair	Tier	Scale	Rot (°)	True (X, Y)	Pred (X, Y)	Error (px)	Status
01	Easy	10.0x	+0.0	(500.0, 500.0)	(500.0, 500.0)	**0.00**	**PASS**
02	Easy	10.0x	+0.0	(480.0, 520.0)	(480.0, 520.0)	**0.00**	**PASS**
03	Easy	9.9x	+0.1	(529.9, 469.9)	(530.0, 469.9)	**0.06**	**PASS**
04	Easy	10.1x	-0.1	(459.9, 539.9)	(460.0, 539.9)	**0.03**	**PASS**
05	Easy	10.0x	+0.0	(510.0, 490.0)	(510.0, 490.0)	**0.00**	**PASS**
06	Scale	9.1x	+0.0	(500.0, 500.0)	(500.0, 500.0)	**0.02**	**PASS**
07	Scale	9.5x	+0.0	(470.0, 530.0)	(470.0, 530.0)	**0.02**	**PASS**
08	Scale	10.0x	+0.0	(520.0, 480.0)	(520.0, 480.0)	**0.00**	**PASS**
09	Scale	10.5x	+0.0	(450.0, 550.0)	(450.0, 550.0)	**0.01**	**PASS**
10	Scale	10.9x	+0.0	(530.0, 460.0)	(530.0, 460.0)	**0.04**	**PASS**
11	Rotation	10.0x	-1.8	(500.0, 500.0)	(500.0, 500.0)	**0.02**	**PASS**
12	Rotation	10.0x	-1.0	(479.6, 519.7)	(479.7, 519.7)	**0.03**	**PASS**
13	Rotation	10.0x	+0.8	(519.7, 479.7)	(519.7, 479.8)	**0.04**	**PASS**
14	Rotation	10.0x	+1.5	(461.1, 541.0)	(461.0, 541.0)	**0.03**	**PASS**
15	Rotation	10.0x	+2.0	(538.6, 458.6)	(538.6, 458.7)	**0.05**	**PASS**
16	SEM Degradation	10.0x	+0.0	(500.0, 500.0)	(500.0, 500.0)	**0.01**	**PASS**
17	SEM Degradation	10.0x	+0.0	(470.0, 530.0)	(470.0, 530.0)	**0.02**	**PASS**
18	SEM Degradation	10.0x	+0.0	(530.0, 470.0)	(530.0, 470.0)	**0.04**	**PASS**
19	SEM Degradation	10.0x	+0.0	(450.0, 550.0)	(450.0, 550.0)	**0.01**	**PASS**
20	SEM Degradation	10.0x	+0.0	(520.0, 480.0)	(520.0, 480.0)	**0.02**	**PASS**
21	Combined	9.2x	+1.6	(354.3, 654.1)	(354.3, 654.2)	**0.04**	**PASS**
22	Combined	9.6x	-1.4	(653.6, 353.7)	(653.6, 353.7)	**0.03**	**PASS**
23	Combined	10.4x	+1.8	(306.4, 706.2)	(306.4, 706.2)	**0.04**	**PASS**
24	Combined	10.8x	-1.7	(705.8, 306.0)	(705.8, 306.1)	**0.03**	**PASS**
25	Combined	9.4x	+1.2	(402.1, 602.1)	(402.1, 602.1)	**0.02**	**PASS**
26	Hard DRAM	9.1x	+1.9	(210.9, 229.4)	(210.6, 220.7)	**8.73**	**FAIL**
27	Hard DRAM	10.9x	-1.9	(770.5, 789.1)	(770.6, 789.1)	**0.08**	**PASS**
28	Hard DRAM	9.2x	-2.0	(241.4, 741.1)	(241.4, 741.1)	**0.05**	**PASS**
29	Hard DRAM	10.8x	+2.0	(741.1, 241.4)	(741.1, 241.4)	**0.03**	**PASS**
30	Hard DRAM	9.1x	+1.9	(210.1, 809.8)	(210.1, 809.8)	**0.04**	**PASS**